



Cubic and quartic net-regular strongly regular signed graphs

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ABSTRACT

In this paper, we determine all connected net-regular strongly regular signed graphs with vertex degree at most four.

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1. Introduction

For a finite ordinary graph $G = (V, E)$, a signed graph $\hat{G}$ is a pair (G, σ) , where $\sigma : E \rightarrow \{+1, -1\}$ is the *sign function* or the *signature*. In this context, G is recognized as the *underlying graph* of $\hat{G}$. The *order* n is the number of vertices of $\hat{G}$. Every edge of $\hat{G}$ is declared positive or negative, and all the positive edges (resp. all the negative edges) induce the subgraph of G denoted by G^+ (resp. G^-). An ordinary graph G is interpreted as a signed graph with the all-positive signature. In our notation it is recognized by the absence of a dot symbol. As usual, isomorphic signed graphs are mutually identified.

The vertex degree in a signed graph coincides with its degree in the corresponding underlying graph. Similarly, we say that a signed graph is connected, complete, regular or bipartite if the same holds for the underlying graph. If a signed graph is regular of degree r , then we simply say that it is r -regular; in particular, it is called *cubic* (resp. *quartic*) if the corresponding degree is three (four). A signed graph is called *homogeneous* if all its edges have the same sign, e.g., if its edge set is empty. Otherwise, it is *inhomogeneous*. The *negation* $-\hat{G}$ of $\hat{G}$ is obtained by reversing the sign of all edges of $\hat{G}$.

A *walk* in a signed graph is defined in the same way as a walk in a graph, i.e., it is a sequence of alternate vertices and edges such that consecutive vertices are incident with the corresponding edge. A walk is *positive* if the number of negative edges contained (counted with their repetition) is even; otherwise, it is *negative*.

The *net degree* of a vertex in $\hat{G}$ is the difference between the number of positive edges and the number of negative edges incident with it. We say that $\hat{G}$ is *net-regular* if the net degree, considered as a function on the vertex set, is constant; in this case, the common net degree is denoted by ϱ . Equivalently, it is a signed graph $\hat{G}$ such that the all-one vector $\mathbf{j}$ is an eigenvector for $A_{\hat{G}}$ (corresponding to the eigenvalue ϱ). Some results on signed graphs that are simultaneously regular and net-regular can be found in [12]. One may observe that a net-regular signed graph does not need to be regular.

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The $n \times n$ adjacency matrix $A_{\hat{G}}$ of $\hat{G}$ is obtained from the standard adjacency matrix of G by reversing the sign of all 1s which correspond to negative edges. By the *eigenvalues* and the *spectrum* of $\hat{G}$, we mean the eigenvalues and the spectrum of $A_{\hat{G}}$. In the entire paper, ‘a signed graph has k eigenvalues’ means ‘a signed graph has exactly k distinct eigenvalues’.

Following [15], we say that a signed graph G is *strongly regular* (for short, $\hat{G}$ is a *SRSG*) whenever it is neither homogeneous complete nor edgeless, and there exist $r \in \mathbb{N}$, $a, b, c \in \mathbb{Z}$, such that the entries of $A_{\hat{G}}^2$ satisfy

$$a_{ij}^{(2)} = \begin{cases} r & \text{if } i = j, \\ a & \text{if } i \overset{+}{\sim} j, \\ b & \text{if } i \overset{-}{\sim} j, \\ c & \text{if } i \not\sim j \text{ and } i \neq j, \end{cases}$$

where $\overset{+}{\sim}$ (resp. $\overset{-}{\sim}$) designates that the corresponding vertices are joined by a positive (negative) edge; similarly, $\not\sim$ stands between non-adjacent vertices.

Obviously, a SRSG $\hat{G}$ is regular with vertex degree r . Observe that $a_{ij}^{(2)}$ is the difference between the number of positive and the number of negative i - j walks of length 2 in $\hat{G}$. Accordingly, this definition generalizes definition of a strongly regular graph in the sense that a strongly regular graph remains strongly regular when it is considered as a signed graph. We know from [15] that every signed graph with two eigenvalues is strongly regular.

Net-regular signed graphs are extensively studied in the recent past. In particular, many characterizations of these structures and particular determinations can be found in [2,11] (where they are considered in the context of products on signed graphs), [6,7] (particular classes including trees), [12] (relations with regular signed graphs), [9] (fixed net degree) or [14] (integral signed graphs). Some notable results on strongly regular signed graphs can be found in [15] (initial results, disconnected and bipartite SRSGs), [18] (general properties), [4] (classification and particular characterizations), [3,5,10,16] (SRSGs with two eigenvalues), [8] (constructions), [17] (relationships with linear ternary codes) or [1] (open problems and related results).

An interplay between net regularity and strong regularity appears sporadically in the previous references, say in [1,9,12,15,17,18]. In this paper, we proceed in the same direction and give the complete classification of net-regular strongly regular signed graphs whose vertex degree does not exceed 4.

The forthcoming sections are organized in the following way. Some notation, terminology and necessary results are given in Section 2. Also, this section reveals all net-regular SRSGs of degree at most 2. Cubic net-regular SRSGs are determined in Section 3, and quartic net-regular SRSGs are determined in Section 4. The main results are reviewed in Section 5.

2. Preliminaries

We use I and J to denote the identity matrix and the all-one matrix, respectively. The size may be written in the subscript.

Switching equivalent signed graphs share the same underlying graph, and one is obtained from the other by taking a subset U of a vertex set V and reversing the sign of every edge with one end in U and the other in $V \setminus U$. Switching equivalent signed graphs share the same spectrum, since the corresponding adjacency matrices are similar. A signed graph is called *balanced* if it switches to its underlying graph; otherwise, it is *unbalanced*. Switching equivalence and balance feature as the fundamental concepts in the theory of signed graphs [18].

One may observe that strong regularity is not an invariant on a switching class of a particular SRSG, unless this SRSG has two eigenvalues (see [15]). In matrix terminology, a signed graph $\hat{G}$ is strongly regular if and only if

$$A_{\hat{G}}^2 + \frac{b-a}{2}A_{\hat{G}} = \frac{a+b}{2}A_G + cA_{\bar{G}} + rI = \frac{a+b-2c}{2}A_G + cJ + (r-c)I. \quad (1)$$

where $\bar{G}$ is the complement of G .

Since a homogeneous SRSG is a strongly regular graph or its negation, in the framework of signed graphs the attention is primarily focused on inhomogeneous ones. If $\hat{G}$ is inhomogeneous, the parameters a and b are uniquely determined. If, in addition, $\hat{G}$ is non-complete then c is fixed, as well, and in this case we usually write the parameters of $\hat{G}$ in the form of the ordered quintuple (n, r, a, b, c) . In [4], all inhomogeneous SRSGs are partitioned into the following 5 disjoint classes:

- C_1 : SRSGs with $a = -b$, which are either complete or non-complete with $c \neq 0$.
- C_2 : SRSGs with $a = -b$, which are non-complete with $c = 0$.
- C_3 : SRSGs with $a \neq -b$, which are either complete or non-complete with $c = \frac{a+b}{2}$.
- C_4 : SRSGs with $a \neq -b$, which are non-complete with $c = 0$.
- C_5 : SRSGs with $a \neq -b$, which are non-complete with $c \neq \frac{a+b}{2}$ and $c \neq 0$.

This classification will be frequently used in the forthcoming sections.

Since the signs of 2-walks are the same in $\hat{G}$ and $-\hat{G}$, the next lemma follows immediately from the definition.

Lemma 2.1. *If $\hat{G}$ is a SRSG belonging to C_i , $1 \leq i \leq 5$, so is $-\hat{G}$. The parameters a and b of an inhomogeneous SRSG $\hat{G}$ interchange their roles in the set of parameters of $-\hat{G}$.*

SRSs we consider are net-regular, so G^+ and G^- are regular graphs, and the net degree ϱ of $\dot{G}$ is the difference between degrees of G^+ and G^- . The previous lemma enables us to restrict our considerations to SRSs with a non-negative net degree. Moreover, we know from [15] that a disconnected inhomogeneous signed graph is strongly regular if and only if every component is a SRS with the same parameters r, a, b and $c = 0$. Thus, we may restrict ourselves to determination of connected SRSs, as the disconnected ones are easily constructed. Accordingly, except in statement formulations, we assume that a signed graph under consideration is connected with a non-negative net degree.

The only 1-regular signed graphs are K_2 and $-K_2$, so they are not strongly regular because both are homogeneous and complete. For 2-regular graphs (cycles), the only signature that provides net-regular ones can be defined on cycles of even length (where negative edges induce a perfect matching), and then the net degree is 0. Obviously, in such a signed cycle $\dot{C}_{2k}$, $k \geq 2$, for the vertices at distance 2 there is one negative walk of length 2 between them, giving $c = -1$. Thus, if the diameter is greater than 2, such a signed graph cannot be strongly regular (vertices at distance greater than 2 have no common neighbours). Thus, the unique net-regular SRS with degree 2 is the 4-vertex cycle in which negative edges induce a perfect matching.

Gathering the previous observations, we arrive at the following result.

Theorem 2.2. *There are no connected 1-regular SRSs. The unique connected 2-regular net-regular SRS is the 4-vertex cycle in which negative edges form a perfect matching.*

3. Cubic net-regular SRSs

In this section we determine all cubic net-regular SRSs. First, we prove a proposition that will be used in this and the forthcoming section.

Proposition 3.1. *Let $\dot{G}$ be a connected non-complete net-regular SRS belonging to $\mathcal{C}_1 \cup \mathcal{C}_4 \cup \mathcal{C}_5$. Then for any two vertices i and j of $\dot{G}$, the number of negative walks of length 2 between i and j is even. Moreover, if this number is $2k$, $k \in \mathbb{N}$, then there are exactly k common neighbours of i and j that are joined to i by a positive edge and to j by a negative edge, and exactly k common neighbours that are joined to i by a negative edge and to j by a positive edge.*

Proof. If $\dot{G} \in \mathcal{C}_1 \cup \mathcal{C}_4 \cup \mathcal{C}_5$ is connected and non-complete, then the coefficient of A_G in (1) does not vanish. Clearly, $A_{\dot{G}}$ commutes with the left-hand side of (1), so it commutes with the right-hand side, as well. In addition, $\dot{G}$ is net-regular, and thus $A_{\dot{G}}$ commutes with J , which implies that $A_{\dot{G}}$ commutes with A_G . This yields that the matrices $A_{G^+} = \frac{1}{2}(A_G + A_{\dot{G}})$ and $A_{G^-} = \frac{1}{2}(A_G - A_{\dot{G}})$ mutually commute.

Now, the number of common neighbours of i and j that are joined to i by a positive edge and to j by a negative edge is the (i, j) -entry of $A_{G^+}A_{G^-}$. Similarly, the remaining number given in the statement formulation is the (i, j) -entry of $A_{G^-}A_{G^+}$. The desired result follows by commutativity of A_{G^+} and A_{G^-} . $\square$

If $\dot{G}$ is a cubic net-regular signed graph with a non-negative net degree, then G^+ and G^- are regular of degree 2 and 1, respectively. Hence, G^- induces a perfect matching in $\dot{G}$.

If either $\dot{G} \in \mathcal{C}_2$, or $\dot{G}$ is a complete SRS from $\mathcal{C}_1$, then the defining Eq. (1) reduces to $A_G^2 + bA_{\dot{G}} = 3I$. Obviously, $\dot{G}$ has two eigenvalues, where one of them is $\varrho = 1$, and so the other one must be -3 (as the product of eigenvalues is the negation of the common vertex degree). Then the eigenvalues of $-\dot{G}$ are 3 and $\varrho = -1$. Now, [14, Lemma 2.1] yields that $-\dot{G}$ is switching equivalent to K_4 , and $\dot{G} \in \mathcal{C}_1$. It is the net-regular SRS $\dot{H}_1$, with parameters $(n, r, a, b) = (4, 3, -2, 2)$, obtained by reversing the sign of a perfect matching in K_4 ; see the forthcoming Fig. 1.

Suppose that $\dot{G} \in \mathcal{C}_3$. We know from [4] that every SRS of $\mathcal{C}_3$ whose degree is at most 6 must be complete. However, it can be easily verified that there is only one signature on K_4 that satisfies above conditions, and it produces already obtained SRS $\dot{H}_1$.

Thus, it remains to consider the case of non-complete SRSs $\dot{G} \in \mathcal{C}_1 \cup \mathcal{C}_4 \cup \mathcal{C}_5$.

Lemma 3.2. *There are no non-complete cubic net-regular (with $\varrho = 1$) SRSs that belong to $\mathcal{C}_1 \cup \mathcal{C}_4 \cup \mathcal{C}_5$ and contain an unbalanced triangle.*

Proof. Let the vertices i, j, k , with $i \overset{+}{\sim} j$, $j \overset{-}{\sim} k$, $k \overset{+}{\sim} i$, form an unbalanced triangle in $\dot{G}$. According to Proposition 3.1, i and j must have a common neighbour, say l , such that $i \overset{-}{\sim} l$ and $j \overset{+}{\sim} l$. By the same proposition, since $k \overset{+}{\sim} i$, there is a positive edge between k and l . Now, all four vertices have degree 3, and the obtained graph is the complete SRS already denoted by $\dot{H}_1$. $\square$

Suppose now that a non-complete cubic net-regular SRS $\dot{G}$ from $\mathcal{C}_1 \cup \mathcal{C}_4 \cup \mathcal{C}_5$ does not contain an unbalanced triangle. Then the parameter b equals 0. Also, there are no negative walks of length 2 between two vertices joined by a positive edge, so a is either 1 or 0. Consider now the defining Eq. (1), and observe that $\mathbf{j}$ is an eigenvector for all four matrices that appear in it (it corresponds to the eigenvalue 1 of A_G^2 and $A_{\dot{G}}$, 3 of A_G and $n - 4$ of $A_{\dot{G}}$). We have $1 = 2a + b + c(n - 4) + 3$. If $a = 1$, $b = 0$, the last equality reduces to $c(n - 4) = -4$, and if $a = 0$, $b = 0$ then it reduces to $c(n - 4) = -2$. Obviously, $c < 0$ holds, and at the end we have exactly four possible quintuples of parameters:

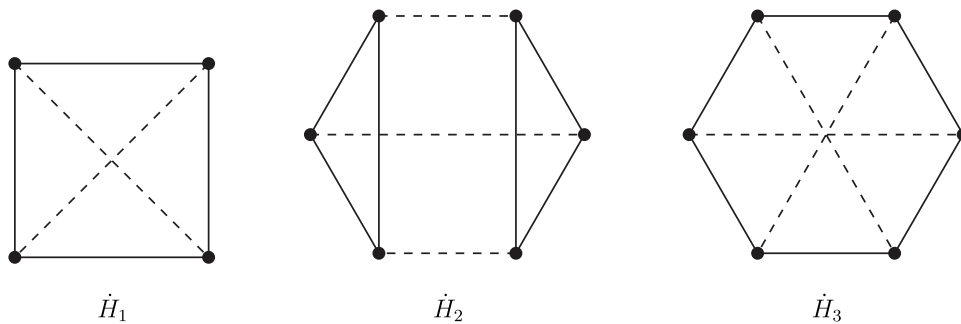


Fig. 1. Connected cubic net-regular SRSs with net degree 1.

- (i) $(8, 3, 1, 0, -1)$;
- (ii) $(6, 3, 1, 0, -2)$;
- (iii) $(6, 3, 0, 0, -1)$;
- (iv) $(5, 3, 0, 0, -2)$.

Consider the first possibility. Since $a = 1$ and $b = 0$, positive edges form a disjoint union of triangles, so G^+ is isomorphic to tK_3 for some positive integer t , and then the order of $\hat{G}$ would have to be divisible by 3. This eliminates the first parameter set.

There are 2 cubic graphs on 6 vertices, one of them contains two triangles, and the other one is the complete bipartite graph $K_{3,3}$. The first one allows a signature which defines a cubic net-regular SRS $\dot{H}_2$, where both triangles are with positive edges, and the other 3 edges, forming a perfect matching, are negative. The second one gives rise to the cubic net-regular SRS $\dot{H}_3$ obtained by reversing the sign of all edges of a perfect matching in $K_{3,3}$. $\dot{H}_2$ and $\dot{H}_3$ are illustrated in Fig. 1.

The fourth possibility cannot produce cubic net-regular SRS, because it does not contain a perfect matching (its order is odd).

We conclude this section with the following theorem that summarizes the previous discussion.

Theorem 3.3. *There are six connected cubic net-regular SRSs. They are: $\dot{H}_1, \dot{H}_2, \dot{H}_3$ (illustrated in Fig. 1), $-\dot{H}_1, -\dot{H}_2$ and $-\dot{H}_3$.*

For the record, $\dot{H}_1, \dot{H}_3$ and their negations belong to $\mathcal{C}_1$, whereas $\dot{H}_2$ and its negation belong to $\mathcal{C}_5$.

4. Quartic net-regular SRSs

For net-regular SRSs of degree four, we have two possibilities for the net degree: $\varrho = 2$ or $\varrho = 0$. We discuss them separately.

4.1. Quartic net-regular SRSs with net degree 2

Let $\hat{G}$ be a signed graph defined in the last title. In this case, G^+ and G^- are regular graphs of degree 3 and 1, respectively. Accordingly, G^- induces a perfect matching in $\hat{G}$.

As before, if $\hat{G} \in \mathcal{C}_3$, then it must be complete. However, since the order of K_5 is odd, $\hat{G}$ does not have a perfect matching. This observation also eliminates the possibility that $\hat{G}$ is a complete SRS of $\mathcal{C}_1$.

Assume now that $\hat{G} \in \mathcal{C}_2$. Then the defining Eq. (1) reads $A_{\hat{G}}^2 + bA_{\hat{G}} = 4I$. Clearly, since $\hat{G}$ has two eigenvalues and one of them is $\varrho = 2$, the other one must be -2 , and consequently $b = 0$. All signed graphs with eigenvalues 2 and -2 are known, and they can be found in any of [3,5,16]. Accordingly, up to switching equivalence, there are two individual signed graphs with this spectral property, usually denoted by $\dot{S}_{14}$ and $\dot{S}_{16}$, along with one infinite family $\dot{T}_{2k}$, $k \geq 3$, consisting of the so-called $2k$ -vertex maximal connected cyclotomic toral tessellations. These signed graphs are illustrated in Fig. 2.

We have already pointed out that every signed graph with two eigenvalues is strongly regular, and so it remains to extract the net-regular ones (of net degree 2). Every such a signed graph switches to some of the mentioned ones. Moreover, signed graphs $\dot{S}_{14}$ and $\dot{S}_{16}$ satisfy the assumption on net-regularity, and by a computer search we have checked that every other net-regular switching is isomorphic to them.

Consider now toral tessellations. First, for $\dot{T}_6$ (the only signed graph in the family which is not bipartite) it can be easily verified, even by hand, that it does not switch to any net-regular signed graph. Next, $\dot{T}_8$ is complete bipartite, and thus the only solution is identified as a signed graph in which a perfect matching corresponds to negative edges.

Assume that $k \geq 5$ and let T_{2k} be the underlying graph of $\dot{T}_{2k}$, along with the vertex labellings as in Fig. 2. We are going to assign a signature to T_{2k} in such a way that the resulting signed graph is net-regular with net degree 2 and the

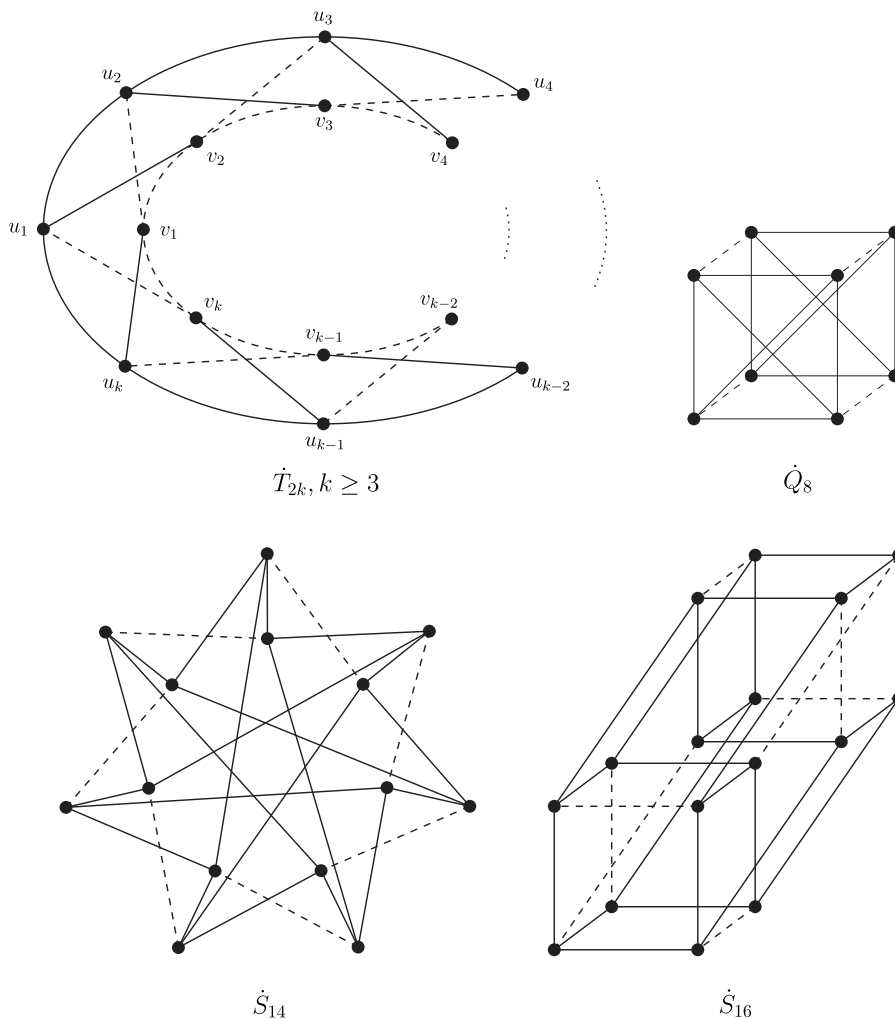


Fig. 2. Signed graphs $\dot{T}_{2k}$, $k \geq 3$, $\dot{Q}_8$, $\dot{S}_{14}$ and $\dot{S}_{16}$.

square of its adjacency matrix is $4I$; this translates to the property that the difference between positive and negative walks (i.e., paths) of length two between two non-adjacent vertices equals zero. Note that non-adjacent vertices at distance 2 in T_{2k} , v_i and u_i , $1 \leq i \leq k$, have four common neighbours, whereas every other pair of vertices at distance 2 has two common neighbours. All other vertices of T_{2k} have no common neighbours. Thus, there must exist two positive and two negative paths of length two between vertices v_i and u_i , $1 \leq i \leq k$, and one positive and one negative path of length two between every other pair of vertices at distance two. This leads to the following lemma.

Lemma 4.1. Let $\dot{G}$ be a net-regular signed graph with net degree 2 and underlying graph T_{2k} , $k \geq 5$. If $A_{\dot{G}}^2 = 4I$, then the following statements hold true:

- (i) If $v_i \sim v_{i+1}$, for some $1 \leq i \leq k-1$, then $u_i \sim u_{i+1}$;
- (ii) If $v_i \sim u_{i+1}$, for some $1 \leq i \leq k-1$, then $u_i \sim v_{i+1}$;
- (iii) If $v_1 \sim v_k$, then $u_1 \sim u_k$;
- (iv) If $v_1 \sim u_k$, then $u_1 \sim v_k$.

Proof. We consider (i). Suppose that $v_i \sim v_{i+1}$ and $i \leq k-2$. There are three possibilities for a negative edge incident with u_{i+1} : $u_{i+1} \sim u_i$, or $u_{i+1} \sim v_{i+2}$, or $u_{i+1} \sim u_{i+2}$. If $i = k-1$, the possibilities are: $u_k \sim u_1$, or $u_k \sim v_1$, or $u_k \sim u_{k-1}$. But if $u_{i+1} \sim v_{i+2}$ (resp. $u_k \sim v_1$), then there would be two negative paths of length 2 between v_i and v_{i+2} (v_1 and v_{i-1}). Similarly, if $u_{i+1} \sim u_{i+2}$ (resp. $u_k \sim u_1$) there would be two negative paths of length 2 between v_i and u_{i+2} (v_{k-1} and u_1). This proves (i).

We proceed with (ii). For $v_i \sim u_{i+1}$ and $i \geq 2$, there are three possibilities for a negative edge incident with u_i : $u_i \sim v_{i+1}$, or $u_i \sim u_{i-1}$, or $u_i \sim v_{i-1}$. If $i = 1$, the possibilities are: $u_1 \sim v_2$, or $u_1 \sim u_k$, or $u_1 \sim v_k$. As before, if $u_i \sim u_{i-1}$ (resp. $u_1 \sim u_k$), then there would be two negative paths of length 2 between u_{i-1} and u_{i+1} (u_k and u_2). Also, if $u_i \sim v_{i-1}$ (resp. $u_1 \sim v_k$), there would be two negative paths of length 2 between vertices u_{i+1} and v_{i-1} (v_k and u_2). Hence, part (ii) follows.

The proofs of the remaining two statements are virtually the same as the proofs of the first two statements. $\square$

Lemma 4.1 yields that $\dot{G}$ has a signature such that negative edges are paired in the following way: If, for some $i \leq k-1$, $v_i \sim v_{i+1}$ then $u_i \sim u_{i+1}$, and if $v_i \sim v_k$ then $u_1 \sim u_k$. Next, if, for some $i \leq k-1$, $v_i \sim u_{i+1}$ then $u_i \sim v_{i+1}$, and if $v_k \sim u_1$ then $u_k \sim v_1$. It immediately follows that, since negative edges form a perfect matching, k must be even. Now, if $\dot{G}$ is underlined by T_{4j} , for $j \geq 3$, and its signature satisfies the previous conditions, then $\dot{G}$ is isomorphic to the signed graph whose negative edges are $v_{2l-1} \sim v_{2l}$, $u_{2l-1} \sim u_{2l}$, where l takes all values ranging from 1 to j . Indeed, the corresponding isomorphism f is given by: $f(v_{2i}) = v_{2i}$, $f(u_{2i}) = u_{2i}$, $f(v_{2i-1}) = v_{2i-1}$, $f(u_{2i-1}) = u_{2i-1}$ if $v_{2i-1} \sim v_{2i}$ and $u_{2i-1} \sim u_{2i}$, whereas $f(v_{2i-1}) = u_{2i-1}$, $f(u_{2i-1}) = v_{2i-1}$ if $v_{2i-1} \sim u_{2i}$ and $u_{2i-1} \sim v_{2i}$, $1 \leq i \leq j$.

We formulate the previous discussion in the following lemma.

Lemma 4.2. Let $\dot{G} \in \mathcal{C}_2$ be a connected quartic net-regular SRSG with net degree 2. Then $\dot{G}$ is isomorphic to either $\dot{S}_{14}$, or $\dot{S}_{16}$, or a switching of $\dot{T}_{4j}$, $j \geq 2$, where in the latter signed graph negative edges are $v_{2i-1}v_{2i}$ and $u_{2i-1}u_{2i}$, for $1 \leq i \leq j$.

Remark 4.3. We note that the signed graph $\dot{T}_{4j}$ of the previous lemma is obtained from $\dot{T}_{2k}$ of Fig. 2 by setting $k = 2j$, interchanging the roles of u_{2i-1} and v_{2i-1} , for $1 \leq i \leq 2j$, and finally making a switch with respect to vertices v_{2i} , $1 \leq i \leq 2j$. In other words, one cycle whose edges alternate in sign is $v_1u_2v_3u_4 \cdots v_{2j-1}u_{2j}v_1$, the other one is $u_1v_2u_3v_4 \cdots u_{2j-1}v_{2j}u_1$, and the edges between these cycles are positive.

Let now $\dot{G} \in \mathcal{C}_1 \cup \mathcal{C}_4 \cup \mathcal{C}_5$. We can also assume that $\dot{G}$ is non-complete, as $\mathcal{C}_4$ and $\mathcal{C}_5$ do not contain complete signed graphs, whereas complete signed graphs of $\mathcal{C}_1$ are eliminated at the beginning of this subsection. We prove a lemma.

Lemma 4.4. Let $\dot{G} \in \mathcal{C}_1 \cup \mathcal{C}_4 \cup \mathcal{C}_5$ be a connected non-complete quartic net-regular SRSG with $q = 2$. Then $b \neq 3$.

Proof. Suppose that $b = 3$ holds. Consider a pair of vertices i, j joined by a negative edge. Since $b = 3$, there are three vertices, say k, l and m such that $j \stackrel{+}{\sim} k$, $k \stackrel{+}{\sim} i$, $j \stackrel{+}{\sim} l$, $l \stackrel{+}{\sim} i$, $j \stackrel{+}{\sim} m$ and $m \stackrel{+}{\sim} i$. Concerning the edges between i, j and k and employing Proposition 3.1, we deduce that there must be either $k \sim l$ or $k \sim m$. Next, concerning edges between i, j and m and taking into account the same proposition, we arrive at $m \sim k$ or $m \sim l$. In the same way, we arrive at $l \sim k$ or $l \sim m$.

Gathering the previous conclusions, we deduce that $k \sim l$ or $k \sim m$ and $m \sim k$ or $m \sim l$ and $l \sim k$ or $l \sim m$, which is impossible, because then either k , or m , or l would be incident with two negative edges. $\square$

We now assume that $\dot{G}$ contains an unbalanced triangle. Then, in the light of Proposition 3.1, possible values of a and b are $a = -2$ or $a = -1$, and $b = 1$, $b = 2$, or $b = 3$. The possibility $b = 3$ is eliminated by Lemma 4.4. From the defining Eq. (1), bearing in mind that $\mathbf{j}$ is an eigenvector of all four matrices that appear in it (corresponding to the eigenvalue 4 of A_G^2 , 2 of A_G , 4 of A_G and $n - 5$ of A_G), we obtain $3a + b + c(n - 5) = 0$. Now we have to examine the following sets of parameters:

- (i) (10, 4, -2, 1, 1);
- (ii) (9, 4, -2, 2, 1);
- (iii) (7, 4, -2, 2, 2);
- (iv) (6, 4, -2, 2, 4);
- (v) (7, 4, -1, 1, 1);
- (vi) (6, 4, -1, 1, 2);
- (vii) (6, 4, -1, 2, 1).

The sets (ii), (iii) and (v) are discarded immediately, since the order of the corresponding signed graphs is odd, so they cannot contain a perfect matching. The sets (iv), (vi) and (vii) all correspond to signed graphs of order 6, and there is only one quartic graph of order 6 – the complete tripartite graph $K_{2,2,2}$. All perfect matchings of $K_{2,2,2}$ produce isomorphic signed graphs which are not strongly regular.

Consider now the remaining parameters (i). Let i and j be vertices of $\dot{G}$ joined by a positive edge. Then, since $a = -2$, there are vertices k and l , such that $i \stackrel{+}{\sim} k$, $i \sim l$, $j \sim k$, $j \stackrel{+}{\sim} l$. Consider the vertices k and l . If they are adjacent, then $k \stackrel{+}{\sim} l$ must hold, but then $b \geq 2$. If they are non-adjacent, then $c \leq 0$. Thus, there is no such graph.

Therefore, $\dot{G}$ does not contain an unbalanced triangle. Suppose now that $\dot{G}$ contains at least one triangle. Then all triangles must be with all three positive edges, and the possible values for a are 2 or 1, and for b only 0. Now, similarly

to above, the defining Eq. (1) yields $c(n-5) = -6$ for $a = 2$, and $c(n-5) = -3$ for $a = 1$. Clearly c must be negative, and its minimal value is -2 , so we get the three parameter sets: $(8, 4, 2, 0, -2)$, $(11, 4, 2, 0, -1)$ and $(8, 4, 1, 0, -1)$. The second one is eliminated since the number of vertices is odd. The third one is also impossible: note that $a = 1$ implies that every positive edge is contained in exactly one balanced triangle. Let i, j and k be vertices of $\dot{G}$ that form a balanced triangle. Since every vertex is incident with three positive edges, there must be a vertex l such that $i \overset{+}{\sim} l$. But then, l must have a common neighbour with i , so either $l \overset{+}{\sim} j$ or $l \overset{+}{\sim} k$, which implies that there is a positive edge contained in two balanced triangles. It is verified by hand that the remaining possibility gives rise to the net-regular SRSG, say $\dot{Q}_8$, with eight vertices and the adjacency matrix

$$A_{\dot{Q}_8} = \begin{pmatrix} J_4 - I_4 & -I_4 \\ -I_4 & J_4 - I_4 \end{pmatrix}.$$

This signed graph is illustrated in Fig. 2, it is switching equivalent to its underlying graph, belongs to $\mathcal{C}_5$ and has the spectrum: $[4, 2, 0^3, (-2)^3]$. An experienced reader will recognize this graph as a Cartesian product of K_4 and a negative edge.

Finally, we suppose that $\dot{G}$ is triangle-free. Then $a = b = 0$, and the defining Eq. (1) gives $c(n-5) = 0$. If $c = 0$, then $\dot{G}$ belongs to $\mathcal{C}_2$ (but not to $\mathcal{C}_1 \cup \mathcal{C}_4 \cup \mathcal{C}_5$). Otherwise, it holds $n = 5$, so $\dot{G}$ must be complete, contrary to our assumption.

We have considered all possibilities and arrived at the following result.

Theorem 4.5. *There are six connected quartic net-regular SRSGs with net degree 2 or -2 ($\dot{S}_{14}$, $\dot{S}_{16}$, $\dot{Q}_8$, illustrated in Fig. 2, and their negations $-\dot{S}_{14}$, $-\dot{S}_{16}$ and $-\dot{Q}_8$) and one infinite family comprised of signed graphs that switch to $\dot{T}_{4j}$, $j \geq 2$, of Fig. 2 and whose negative edges are $v_{2i-1} \overset{-}{\sim} v_{2i}$ and $u_{2i-1} \overset{-}{\sim} u_{2i}$, for $1 \leq i \leq j$.*

Note that $\dot{Q}_8$ and its negation belong to $\mathcal{C}_5$, while the remaining signed graphs of the previous theorem belong to $\mathcal{C}_2$ (see Lemma 4.2).

4.2. Quartic net-regular SRSGs with net degree 0

In this case, G^+ and G^- are regular graphs of degree 2, so each of them is either a cycle or a disjoint union of cycles.

Note that, since one of eigenvalues of $\dot{G}$ is $\varrho = 0$, it cannot belong to $\mathcal{C}_2$. If $\dot{G}$ is a complete signed graph $\dot{K}_5$, then G^+ and G^- are both isomorphic to C_5 , and then $\dot{G}$ has parameters $(n, r, a, b) = (5, 4, -1, -1)$ and belongs to $\mathcal{C}_3$. In the forthcoming Fig. 3, it is denoted by $\dot{R}_5$; the spectrum is $[\sqrt{5}^2, 0, (-\sqrt{5})^2]$. We have already emphasized that every SRSG of $\mathcal{C}_3$ having degree at most 6 must be complete, and thus $\dot{R}_5$ is the unique solution from this class.

It remains to consider connected and non-complete SRSGs of $\mathcal{C}_1 \cup \mathcal{C}_4 \cup \mathcal{C}_5$.

Lemma 4.6. *Let $\dot{G} \in \mathcal{C}_1 \cup \mathcal{C}_4 \cup \mathcal{C}_5$ be a connected non-complete quartic net-regular SRSG with $\varrho = 0$. The following statements hold true:*

- (i) $a < 3$;
- (ii) If $a = 2$, then $b \leq -1$;
- (iii) If $a = -2$, then $b \geq -1$;
- (iv) If $b = -2$, then $a \geq 1$;
- (v) $a = 0$ (resp. $b = 0$) holds only if vertices joined by a positive (negative) edge have no common neighbours.

Proof. (i): Suppose that $a = 3$ and let i and j make a pair of vertices joined by a positive edge. Then there are vertices k, l and m , such that $i \overset{+}{\sim} k, i \overset{-}{\sim} l, i \overset{-}{\sim} m$ and $j \overset{+}{\sim} k, j \overset{-}{\sim} l, j \overset{-}{\sim} m$. But then, because i and k are joined by a positive edge, $k \overset{-}{\sim} l$ must hold, which is impossible since then l would be incident with at least 3 negative edges.

(ii): If $a = 2$, then there is a negative edge located in a balanced triangle, so there are two vertices of $\dot{G}$ joined by a negative edge, along with one negative walk of length 2 between them. According to Proposition 3.1, there must be another one such walk. Note that there can be at most three walks of length 2 between vertices joined by an edge (as the degree of $\dot{G}$ is 4), so b cannot be greater than -1 .

(iii): If $a = -2$, then there is a negative edge located in an unbalanced triangle, so there are two vertices joined by a negative edge, along with one positive walk of length 2 between them. Since there can be at most three walks of length 2 between vertices joined by an edge in $\dot{G}$, in the light of Proposition 3.1, we conclude that b cannot be less than -1 .

(iv): By Proposition 3.1, if $b = -2$ then there are no positive walks of length 2 between vertices joined by a negative edge. Let i and j be such vertices. It follows that there are vertices k and l such that $i \overset{+}{\sim} k, i \overset{-}{\sim} l$ and $j \overset{+}{\sim} k, j \overset{+}{\sim} l$. Note that k and l cannot be joined by a positive edge; otherwise, there would be a positive walk of length 2 between i and l . But then i and k cannot have negative walks of length 2 between them, and since there is one positive walk of length 2 between them (as $i \overset{-}{\sim} j$ and $k \overset{-}{\sim} j$), we conclude that $a \geq 1$ holds.

(v): This follows from the fact that there can be at most three walks of length 2 between vertices joined by an edge, the fact that the number of negative walks of length 2 must be even (see Proposition 3.1) and the assumption that degree of $\dot{G}$ is 4. $\square$

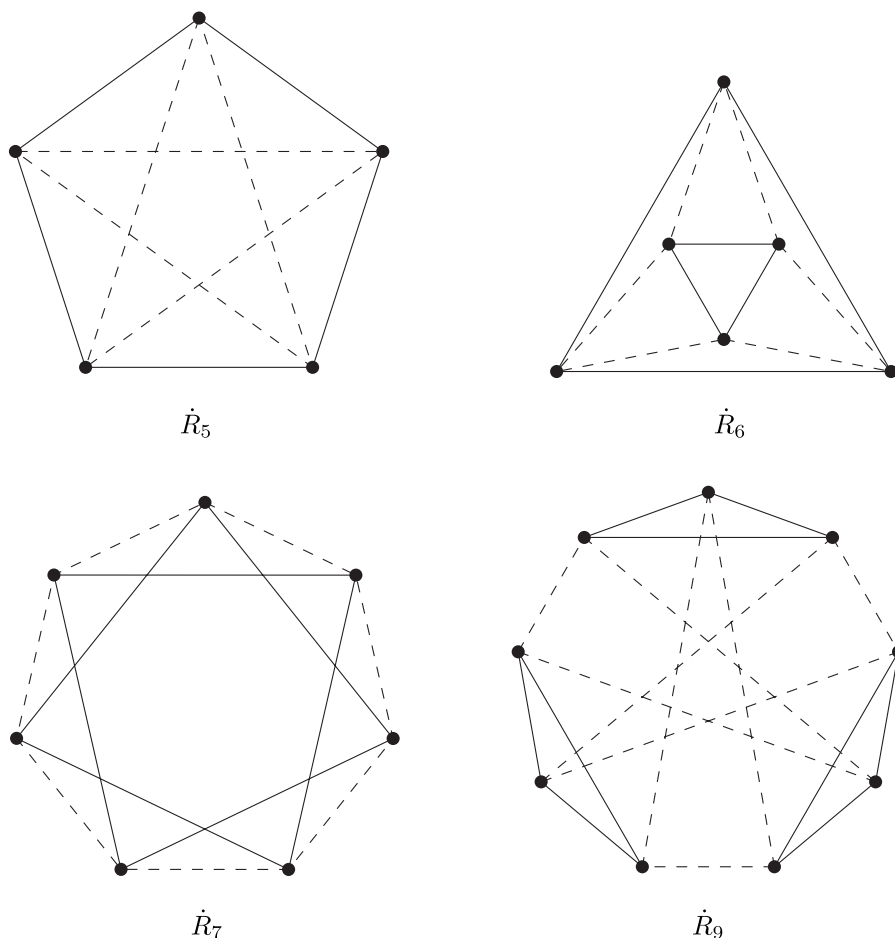


Fig. 3. Connected quartic net-regular SRSs with net degree 0 and $a \geq b$.

By virtue of Lemma 2.1, we may consider only SRSs with $a \geq b$. This restriction and Lemma 4.6, lead to the following possibilities for a and b :

$$(a, b) \in \{(2, -1), (2, -2), (1, 1), (1, 0), (1, -1), (1, -2), (0, 0), (0, -1), (-1, -1)\}.$$

We first eliminate the possibility $b = a = -1$.

Lemma 4.7. If $\dot{G} \in \mathcal{C}_1 \cup \mathcal{C}_4 \cup \mathcal{C}_5$ is a connected non-complete quartic net-regular SRS with $q = 0$, then $b \neq -1$ and $a \neq -1$.

Proof. Let $\dot{G}$ be as in the statement formulation, and assume that $b = -1$ holds. Observe that, according to Proposition 3.1, vertices joined by a negative edge must have all three neighbours in common. Let i and j be such vertices. Then there are vertices k, l, m , such that $i \overset{+}{\sim} k, i \overset{+}{\sim} l, i \overset{+}{\sim} m$ and $j \overset{+}{\sim} k, j \overset{+}{\sim} l, j \overset{+}{\sim} m$. But then, l must be adjacent to each k and m , and also k must be adjacent to each l and m . It follows that $\dot{G}$ is complete, which contradicts the assumption of the statement.

The case $a = -1$ is considered analogously. $\square$

Consider now the defining Eq. (1) and note that $\mathbf{j}$ is an eigenvector for all four matrices that appear in it (corresponding to 0 of A_G^2 and $A_{\dot{G}}$, 4 of A_G and $n - 5$ of $A_{\dot{G}}$). We compute $2a + 2b + c(n - 5) + 4 = 0$.

Lemma 4.8. The signed graph $\dot{R}_6$ of Fig. 3 is the unique connected non-complete quartic net-regular SRS that belongs to $\mathcal{C}_1 \cup \mathcal{C}_4 \cup \mathcal{C}_5$ and satisfies $(q, a, b) = (0, 2, -2)$.

Proof. Let $\dot{G}$ be a SRS satisfying the assumptions of the statement. From the equality given upon the statement formulation, we obtain either $c = -1$ and $n = 9$, or $c = -2$ and $n = 7$, or $c = -4$ and $n = 6$. Since $a = 2$ and $b = -2$, two adjacent vertices in the underlying graph G must have two common neighbours.

Suppose that $c = -1$. Then non-adjacent vertices in G have three common neighbours, so G is strongly regular with parameters $(n, r, a, c) = (9, 4, 2, 3)$. Such strongly regular graph does not exist, which can be seen either by observing tables of small strongly regular graphs (see [13, Subsection 3.4.2]) or by employing the equality

$$(n - r - 1)c = r(r - a - 1) \quad (2)$$

(see [13, p. 72]). Let now $c = -2$. In this case, non-adjacent vertices have two common neighbours, so G is strongly regular with $(n, r, a, c) = (7, 4, 2, 2)$, but such strongly regular graph does not exist (one may observe that the condition that eigenvalue multiplicities are integers is not satisfied).

The last possibility for c induces a strongly regular graph with $(n, r, a, c) = (6, 4, 2, 4)$, i.e., the complete tripartite graph $K_{2,2,2}$. It can be verified by hand that, up to isomorphism, there is exactly one signature defined on this graph that produces the desired SRSg, $\dot{R}_6$ of Fig. 3. $\square$

Next, we consider the case $a = b = 1$.

Lemma 4.9. *The signed graph $\dot{R}_9$ of Fig. 3 is the unique connected non-complete quartic net-regular SRSg that belongs to $\mathcal{C}_1 \cup \mathcal{C}_4 \cup \mathcal{C}_5$ and satisfies $(\varrho, a, b) = (0, 1, 1)$.*

Proof. As in the previous proof, we obtain three possible parameter sets: $(13, 4, 1, 1, -1)$, $(9, 4, 1, 1, -2)$ and $(7, 4, 1, 1, -4)$.

The first one is eliminated since the underlying graph would be strongly regular with $(n, r, a, c) = (13, 4, 1, 3)$, which is impossible as the equality (2) is not satisfied.

The third one, similarly, implies the existence of a strongly regular graph with parameters $(7, 4, 1, 4)$, but it does not exist since the eigenvalue multiplicities are not integers.

Consider the second parameter set. Here, the underlying graph is strongly regular with parameters $(9, 4, 1, 2)$, i.e., it is the Paley graph with 9 vertices. Since $a = b = 1$, positive (and negative) edges form a disjoint union of triangles, so both G^+ and G^- are isomorphic to $3K_3$. Again, there is only one signature realizing this condition, and it produces $\dot{R}_9$. $\square$

We now eliminate the possibility $b = 0$.

Lemma 4.10. *If $\dot{G} \in \mathcal{C}_1 \cup \mathcal{C}_4 \cup \mathcal{C}_5$ is a connected non-complete quartic net-regular SRSg with $\varrho = 0$, then $(a, b) \notin \{(1, 0), (0, 0)\}$.*

Proof. Assume that $a = 1$ and $b = 0$. Then $2a + 2b + c(n - 5) + 4 = 0$ gives either $c = -2$ and $n = 8$, or $c = -1$ and $n = 11$. In this case, according to Lemma 4.6(v), vertices joined by a negative edge have no common neighbours, so positive edges form a disjoint union of triangles, which is impossible because the order is not a multiple of 3.

Let now $a = b = 0$. The same equation now gives either $c = -1$ and $n = 9$, or $c = -2$ and $n = 7$, or $c = -4$ and $n = 6$. Due to Lemma 4.6(v), the underlying graph is strongly regular and without triangles, but there is no such strongly regular graph with 7 or 9 vertices. If the order is 6, the only strongly regular graph without triangles is the complete bipartite $K_{3,3}$, but its degree is 3. $\square$

The remaining possibility is $(a, b) = (1, -2)$. The equation on parameters produces two possible sets: $(7, 4, 1, -2, -1)$ and $(6, 4, 1, -2, -2)$. Since there are only two quartic graphs of order 7 and only one of order 6, we apply a computer search to establish whether we can assign a signature to any of these three graphs to obtain a strongly regular signed graph with given parameters and net degree 0. As a result, exactly one SRSg corresponds to the former parameter set ($\dot{R}_7$ of Fig. 3), and there is no SRSg that corresponds to the latter set.

Gathering the previous results, we arrive at the following theorem.

Theorem 4.11. *There are exactly six connected quartic net-regular SRSgs with net degree 0: $\dot{R}_5$, $\dot{R}_6$, $-\dot{R}_6$, $\dot{R}_7$, $-\dot{R}_7$ and $\dot{R}_9$.*

Observe that $\dot{R}_5$ and $\dot{R}_9$ are isomorphic to their negations. In addition, $\dot{R}_5 \in \mathcal{C}_2$, $\dot{R}_6 \in \mathcal{C}_1$ and $\dot{R}_7, \dot{R}_9 \in \mathcal{C}_5$.

5. Recapitulation

Taking into account the results reported in the previous sections, we arrive at the following conclusion:

- There are no connected 1-regular net-regular SRSgs (Theorem 2.2).
- The 4-vertex cycle with two negative non-adjacent edges is the unique connected 2-regular net-regular SRSg (Theorem 2.2).
- There are exactly six connected cubic net-regular SRSgs (Theorem 3.3).
- There is one infinite family and additional six connected quartic net-regular SRSgs with net degree $\neq 0$ (Theorem 4.5).
- There are exactly six connected quartic net-regular SRSgs with net degree 0 (Theorem 4.11).

In total, there is an infinite family and additional 19 connected net-regular SRSgs with vertex degree at most 4.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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